

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the AI-Driven Emotion Detection & Personalized Learning Support Platform behaves under expected and peak load conditions. This section documents the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing, Stress Testing
Target Module / API	Streamlit App – Emotion Analysis & Response Generation Flow
Test Environment	Local Streamlit Server (localhost:8501)
Test Date	20 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Analyze a short learner problem description (e.g., "I'm lost on recursion")	10	60	Emotion classified successfully
2	Analyze longer, detailed problem descriptions with multiple sentences	20	60	System handles concurrent requests without failure
3	Multiple learners submitting inputs simultaneously	30	120	Response time remains acceptable
4	Continuous submissions to the Streamlit emotion analysis flow	50	180	App remains stable with minimal errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.5 seconds	Pass	Delay due to Gemini API call
3	Throughput (Req/sec)	10 Req/sec	11 Req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.5%	Pass	Few timeout errors observed
5	CPU Utilization	< 80%	70%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	60%	Pass	Memory usage within limits

Step 4: Observations & Analysis

- The Streamlit app successfully handled concurrent user sessions without significant degradation.
- Average response time remained below the target threshold.
- Gemini API and model inference calls introduced slight latency during peak load but remained within acceptable limits.
- No major app crashes or failures were observed during testing.
- CPU and memory consumption stayed within safe operational limits.
- Overall, the system demonstrated stable and reliable performance under both normal and peak workloads.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results here (e.g., Postman Collection Runner summary, response time graphs, server logs during load test).